

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I Nirmal Kumar / 192412003 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

Answer

Functional Blocks of a Machine Learning System

A **Machine Learning (ML) system** is a computational system that learns useful patterns and relationships from data and uses the learned knowledge to make predictions, classifications, or discover hidden structures. The major functional blocks of a machine learning system are **data collection, data preprocessing, feature extraction, model training, and model evaluation**.

1. Data Collection

Data collection is the first stage of a machine learning system. Relevant data is collected from different sources such as **databases, sensors, websites, files, cameras, transaction records, and user applications**.

The quality and quantity of collected data have a major effect on the performance of the machine learning model.

Example:

For a house-price prediction system, data may contain:

- Area of the house
- Number of bedrooms
- Location
- Age of the building
- Number of parking spaces
- Selling price

2. Data Preprocessing

Raw data may contain **missing values, duplicate records, noise, inconsistent formats, and outliers**. Therefore, preprocessing is performed before giving the data to the learning algorithm.

Important preprocessing operations include:

- Handling missing values
- Removing duplicate records
- Removing or handling noisy data
- Detecting outliers
- Data normalization or standardization
- Encoding categorical variables
- Splitting data into training and testing sets

Example:

If the age of a house is missing in some records, the missing value can be replaced using an appropriate statistical method such as the mean or median.

3. Feature Extraction

A **feature** is a measurable property or characteristic of the input data. Feature extraction identifies or generates the most useful information from the preprocessed data.

The purpose of feature extraction is to:

- Reduce unnecessary information
- Reduce dimensionality
- Improve computational efficiency
- Improve model performance
- Identify important characteristics of the data

Example:

In an image classification system, useful features may include **edges, shapes, texture, and pixel characteristics**.

For house-price prediction, features may include **area, number of rooms, location, and building age**.

4. Supervised Learning

In **supervised learning**, the model is trained using **labelled data**. Each input sample is associated with a known target or output value.

The learning system learns a relationship between the input features and their corresponding labels.

Example – Email Spam Classification

A dataset may contain:

Email Features		Label
Contains promotional words		Spam
Contains suspicious links		Spam
Normal business message		Not Spam
Personal communication		Not Spam

The model learns from these labelled examples and predicts whether a new email is **Spam** or **Not Spam**.

Common supervised learning algorithms include:

- Linear Regression
- Logistic Regression
- Decision Tree
- K-Nearest Neighbours
- Support Vector Machine
- Artificial Neural Networks

5. Unsupervised Learning

In **unsupervised learning**, the training data does **not contain predefined labels**. The model attempts to discover hidden patterns, structures, or groups within the data.

Example – Customer Segmentation

A company may have customer information such as:

- Age
- Income
- Purchase frequency
- Total spending

There are no predefined customer categories. A clustering algorithm such as **K-Means** can automatically group customers with similar characteristics.

For example:

Common unsupervised learning techniques include:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)
- Association Rule Learning

6. Model Training

Model training is the stage where the machine learning algorithm learns patterns from the processed features.

In **supervised learning**, the model compares its predicted output with the known label and adjusts its parameters to reduce the prediction error.

In **unsupervised learning**, the algorithm identifies similarities, differences, clusters, or other hidden structures without using predefined labels.

For example, in a neural network, the training process adjusts **weights and biases** to minimize the error between predicted and expected outputs.

7. Model Evaluation

After training, the model must be evaluated to determine how well it performs on unseen or previously unused data.

Different evaluation measures are used depending on the type of problem.

For Classification

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

For Regression

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

For Clustering

- Silhouette Score
- Within-Cluster Sum of Squares (WCSS)
- Cluster quality and separation

Conclusion

A machine learning system consists of a sequence of functional blocks that transform **raw data into useful knowledge**. Data is first collected and preprocessed, followed by extraction or selection of important features. In **supervised learning**, labelled data is used to learn a mapping between inputs and known outputs, whereas **unsupervised learning** uses unlabelled data to discover hidden patterns or groups. Finally, the trained model is evaluated using suitable performance measures. Thus, each functional block plays an important role in developing an accurate and reliable machine learning system.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

Answer

A machine learning-based disaster management system can be designed to **predict and detect disasters such as floods and earthquakes** by combining historical disaster records with real-time data from sensors, weather stations, satellites, and seismic monitoring systems. The system can provide early warnings and support authorities in making timely decisions.

Model Selection

A **neural network, particularly a deep neural network (DNN), LSTM, or CNN-LSTM**, is suitable for disaster prediction because disaster-related data can contain complex and nonlinear relationships.

- **LSTM (Long Short-Term Memory)** is suitable for time-series data such as rainfall, river level, temperature, and seismic activity because it can learn patterns over time.
- **CNN (Convolutional Neural Network)** can be used when satellite or geographical images are available.
- A **CNN-LSTM hybrid model** can combine spatial information from images with temporal information from sensor data.

For a simpler implementation, algorithms such as **Random Forest, Decision Tree, or XGBoost** can also be used, particularly when the dataset contains structured numerical features.

Therefore, an **LSTM or CNN-LSTM model** is appropriate when sufficient historical and real-time sequential data are available.

Data Preprocessing

The collected data may contain missing values, noise, duplicate records, and different measurement scales. The following preprocessing steps are required:

1. Remove duplicate and inconsistent records.
2. Handle missing values using interpolation, mean, median, or suitable estimation techniques.
3. Remove or manage abnormal and noisy sensor readings.
4. Normalize or standardize numerical features.
5. Convert timestamps into a common format.
6. Synchronize data obtained from different sensors and sources.
7. Divide the dataset into training, validation, and testing sets.

4. Feature Selection

Important features should be selected to improve prediction accuracy and reduce unnecessary computation.

For **flood prediction**, useful features include:

- Rainfall intensity and duration
- River and reservoir water levels
- Soil moisture
- Temperature
- Humidity
- Drainage and geographical information
- Historical flood occurrence

For **earthquake-related prediction/risk assessment**, useful features include:

- Seismic wave measurements
- Magnitude
- Depth
- Frequency of seismic events
- Ground acceleration
- Historical earthquake activity
- Distance from fault zones

Feature selection methods such as **correlation analysis**, **mutual information**, **feature importance**, **Recursive Feature Elimination (RFE)**, and **Principal Component Analysis (PCA)** can be used.

5. Training and Validation

The historical dataset is divided into:

- **Training set:** Used to learn model parameters.
- **Validation set:** Used to tune hyperparameters and prevent overfitting.
- **Testing set:** Used only for final performance evaluation.

During training, the model learns the relationship between disaster-related features and the target output, such as **flood/no flood, earthquake risk level, or predicted water level**.

For time-series data, the data should preferably be divided chronologically rather than randomly to avoid using future information during training.

6. Model Evaluation

The trained model is evaluated using suitable performance measures.

For classification:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

For numerical prediction:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

For disaster management, **recall is particularly important**, because failing to identify an actual disaster can have serious consequences.

7. Role in Disaster Management

The proposed system can assist authorities in several ways:

Early Warning:

Real-time sensor data can be continuously analyzed to identify abnormal patterns and issue warnings before the disaster reaches critical levels.

Risk Assessment:

The system can generate risk levels for different geographical regions and identify areas that require immediate attention.

Emergency Response:

Authorities can use predicted risk information to plan evacuation routes, deploy rescue teams, allocate emergency resources, and identify vulnerable locations.

Conclusion

A machine learning-based disaster management system combines **historical and real-time data, preprocessing, feature selection, model training, validation, and evaluation** to predict disaster risks. An **LSTM or CNN-LSTM model** is suitable for complex time-dependent and spatial disaster data, while models such as Random Forest or XGBoost can be effective for structured datasets. Such a system can provide timely **early warnings, risk assessment, and decision support**, thereby improving disaster preparedness and emergency response.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

Answer

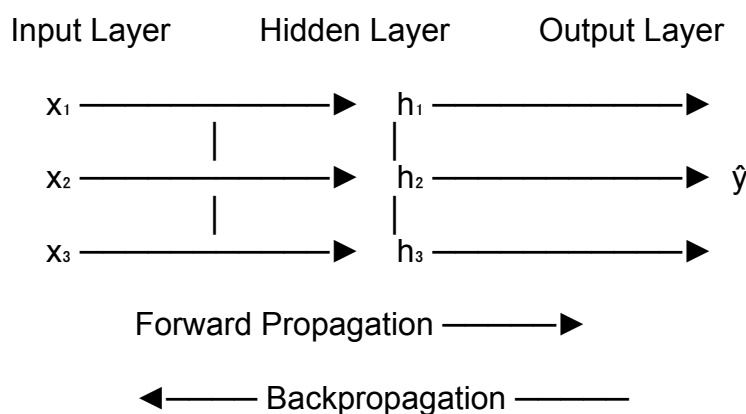
Backpropagation in a Multilayer Perceptron

A **Multilayer Perceptron (MLP)** is a feed-forward neural network consisting of an input layer, one or more hidden layers, and an output layer. For binary classification, the output neuron generally uses a **Sigmoid activation function** to produce an output between 0 and 1.

The **Backpropagation algorithm** is used to train the network by calculating the error at the output and propagating this error backward through the network to update the weights and biases.

1. Structure of the Network

A simple MLP can be represented as:



The network contains weights and biases between the layers. During training, these parameters are adjusted to minimize the difference between the predicted output and the actual target.

2. Forward Pass

In the forward pass, the input data moves from the input layer toward the output layer.

For a neuron in the hidden layer:

$$[z_h = \sum_{i=1}^n w_i x_i + b]$$

The activation is then calculated as:

$$[h = f(z_h)]$$

For a sigmoid activation function:

$$[f(z) = \frac{1}{1 + e^{-z}}]$$

The output neuron calculates:

$$[z_o = \sum w_{hh} + b_o]$$

and

$$[\hat{y} = \frac{1}{1 + e^{-z_o}}]$$

where ($\hat{y}$) is the predicted output.

Simple Example

Assume the network receives:

$$[x_1=1, \quad x_2=0]$$

Suppose the hidden neuron has:

$$[w_1=0.5, \quad w_2=0.4, \quad b=0.1]$$

Therefore,

$$[z_h = (1)(0.5) + (0)(0.4) + 0.1 = 0.6]$$

Using the sigmoid function:

$$h = \frac{1}{1 + e^{-0.6}} \approx 0.646$$

Suppose the output neuron has:

$$w_o = 0.7, \quad b_o = 0.1$$

Then:

$$z_o = (0.646)(0.7) + 0.1 = 0.552$$

Therefore,

$$\hat{y} = \frac{1}{1 + e^{-0.552}} \approx 0.635$$

If the actual target is:

$$y = 1$$

the network has produced an output of approximately **0.635**, which indicates that the prediction is not yet perfect.

3. Error Calculation

For binary classification, binary cross-entropy loss can be used:

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

For the above example:

$$L = -\log(0.635) \approx 0.454$$

This error indicates how far the predicted output is from the actual target.

4. Backward Propagation of Error

In backpropagation, the error is propagated from the output layer toward the input layer using the **chain rule of differentiation**.

For a sigmoid output with binary cross-entropy loss:

$$\begin{aligned} &[\delta_o = \hat{y} - y] \end{aligned}$$

Therefore,

$$\begin{aligned} &[\delta_o = 0.635 - 1 = -0.365] \end{aligned}$$

The output-layer weight gradient is:

$$\begin{aligned} &[\frac{\partial L}{\partial w_o} = \delta_o h] \end{aligned}$$

$$\begin{aligned} &[\frac{\partial L}{\partial w_o} = (-0.365)(0.646) \approx -0.236] \end{aligned}$$

The weight is updated using:

$$\begin{aligned} &[w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}] \end{aligned}$$

where (η) is the learning rate.

Assume:

$$\begin{aligned} &[\eta = 0.1] \end{aligned}$$

Then:

$$\begin{aligned} &[w_{o,\text{new}} = 0.7 - (0.1)(-0.236) \end{aligned}$$

$$\begin{aligned} &[w_{o,\text{new}} \approx 0.724] \end{aligned}$$

Thus, the output weight changes from **0.700 to approximately 0.724**.

The same process is continued backward through the hidden layer, where the error signal is calculated using the downstream error, corresponding weights, and the derivative of the hidden-layer activation function. All weights and biases are updated in this manner.

5. Effect of Learning Rate

The **learning rate (η)** determines the size of each weight update.

- **Very small learning rate:** Training is stable but convergence is slow.
- **Very large learning rate:** Training may overshoot the minimum, oscillate, or fail to converge.
- **Appropriate learning rate:** Provides faster and stable convergence toward a minimum loss.

Therefore, selecting a suitable learning rate is important for achieving good model performance.

6. Effect of Activation Functions

Activation functions introduce **non-linearity** into neural networks and strongly influence training.

Activation Function	Characteristics
Sigmoid	Produces values between 0 and 1; useful for binary output but can suffer from vanishing gradients.
Tanh	Produces values between -1 and 1 and is zero-centered, but may also suffer from vanishing gradients.
ReLU	Computationally simple and generally provides faster training in hidden layers.
Leaky ReLU	Reduces the dead-neuron problem associated with standard ReLU.

For binary classification, **Sigmoid is commonly used in the output layer**, while **ReLU or its variants are commonly used in hidden layers**.

Conclusion

Backpropagation trains an MLP by performing a **forward pass, calculating the prediction error, propagating the error backward, calculating gradients, and updating weights and biases**. The learning rate controls the magnitude of parameter updates, while the choice of activation function affects gradient flow, convergence speed, and prediction accuracy. Proper selection of both is essential for effective neural network training.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

Answer

Genetic Algorithm for Optimizing Speed and Accuracy

A **Genetic Algorithm (GA)** is an evolutionary optimization technique inspired by natural selection. It searches for an optimal solution by repeatedly applying **selection, crossover, and mutation** to a population of candidate solutions.

Given dataset:

Example	Speed	Accuracy	Fitness Value	Suitable
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

The objective is to **maximize the fitness value** while obtaining a suitable combination of Speed and Accuracy.

1. Chromosome Representation

Each candidate solution can be represented as a chromosome containing the two parameters:

Chromosome = [Speed | Accuracy]

For illustration, the parameters can be encoded as:

Low = 00
Medium = 01
High = 10

Therefore, the initial population is:

Example	Speed	Accuracy	Chromosome	Fitness
1	High	Low	10 00	40
2	Medium	Medium	01 01	60
3	High	Medium	10 01	75
4	Low	High	00 10	65
5	Medium	High	01 10	80

2. Initial Population

The initial population contains five candidate solutions:

$P_0 = \{\text{Example 1, Example 2, Example 3, Example 4, Example 5}\}$

Their fitness values are:

40, 60, 75, 65, 80

The highest fitness in the initial population is:

[
Fitness_{max}=80
]

Therefore, **Example 5 is initially the best candidate.**

3. Selection

Selection chooses individuals with better fitness values so that they have a higher probability of producing the next generation.

The fitness ranking is:

Example 5 → 80 → Best
Example 3 → 75
Example 4 → 65
Example 2 → 60
Example 1 → 40 → Weakest

Thus, Examples **5, 3, and 4** are preferred for reproduction.

The purpose of selection is to give better solutions a greater chance of contributing their characteristics to the next generation.

4. Crossover

Crossover combines genetic information from two selected parents to generate new offspring.

Consider:

Parent 1 = Example 5 = 01 | 10
Parent 2 = Example 3 = 10 | 01

Using a single-point crossover between Speed and Accuracy:

Parent 1 = 01 | 10
Parent 2 = 10 | 01

↑
Crossover

The offspring can be:

Offspring 1 = 01 | 01
Offspring 2 = 10 | 10

The second offspring represents:

Speed = High
Accuracy = High

This demonstrates how crossover can combine useful characteristics from different parents.

Important: The fitness of a newly generated chromosome must be calculated using the actual fitness function. The given dataset does not provide a formula for calculating the fitness of unseen combinations.

5. Mutation

Mutation randomly changes one or more genes in a chromosome. It helps the algorithm explore new solutions and prevents the population from becoming too similar.

For example:

Before mutation:

01 | 01
Medium | Medium

After mutation:

01 | 10
Medium | High

Thus, mutation can introduce a new candidate solution that was not present in the original population.

Mutation helps the GA to:

- Explore new regions of the hypothesis space.
- Avoid premature convergence.
- Maintain population diversity.
- Increase the possibility of finding a better solution.

6. Evolution Across Generations

The evolution can be represented as follows:

Generation 0 – Initial Population

Example 1 → Fitness 40

Example 2 → Fitness 60

Example 3 → Fitness 75

Example 4 → Fitness 65

Example 5 → Fitness 80 ← Best

Generation 1 – Selection and Reproduction

The stronger candidates, particularly Examples 5, 3, and 4, are selected.

Selected Parents:

Example 5 → 80

Example 3 → 75

Example 4 → 65

Crossover combines their genetic characteristics to generate offspring.

Parent 5 + Parent 3



Crossover



New Candidate(s)

Mutation may then modify one parameter to introduce additional diversity.

Generation 2 – Evaluation

The newly generated candidates are evaluated using the fitness function. Candidates with higher fitness are retained for the next generation, while weaker candidates have a lower probability of survival.

Higher Fitness



Selection



Crossover



Mutation



New Population



Fitness Evaluation



Repeat

The process continues until the population converges or a satisfactory fitness level is reached.

7. Best Optimal Solution

From the **given dataset**, the highest fitness value is:

```
[  
  \boxed{Fitness=80}  
]
```

Therefore, the best observed solution is:

```
[  
  \boxed{\text{Speed = Medium, Accuracy = High}}  
]
```

with:

```
[  
  \boxed{\text{Fitness = 80, Suitable = Yes}}  
]
```

This corresponds to **Example 5**.

If the GA generates a new combination such as **High Speed + High Accuracy**, that chromosome cannot be declared the optimal solution from the supplied data alone. Its fitness must first be calculated using the specified fitness function.

Conclusion

The Genetic Algorithm improves candidate solutions through repeated **selection, crossover, and mutation**. Selection favors high-fitness individuals, crossover combines useful characteristics of selected parents, and mutation explores new regions of the hypothesis space. For the given dataset, **Example 5 (Medium Speed, High Accuracy)** has the highest known fitness of **80** and is therefore the final optimal solution based on the available data.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network

model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

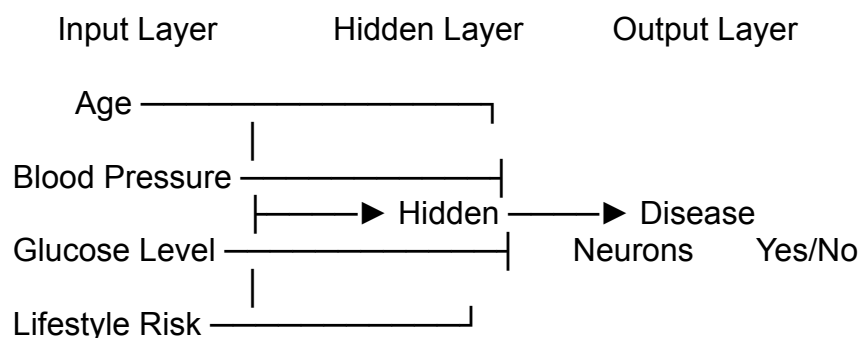
Answer

Neural Network Classification Using Backpropagation

A **Neural Network** can be used to classify whether a patient has a disease based on attributes such as **Age**, **Blood Pressure**, **Glucose Level**, and **Lifestyle Risk**. The output is binary:

- **Yes (1)** → Disease Present
- **No (0)** → Disease Absent

A simple neural network consists of an **input layer**, **hidden layer**, and **output layer**.



1. Training Using Backpropagation

The neural network is trained using the following steps:

Step 1: Forward Propagation

The input features are passed through the network. Each neuron calculates a weighted sum:

$$z = \sum w_i x_i + b$$

An activation function is then applied:

$$a = f(z)$$

For binary classification, a **Sigmoid activation function** is commonly used in the output layer:

$$f(z) = \frac{1}{1 + e^{-z}}$$

The output value lies between 0 and 1. A threshold can be used to classify the patient:

$$\hat{y} = \begin{cases} \text{Yes, } & \hat{y} \geq 0.5 \\ \text{No, } & \hat{y} < 0.5 \end{cases}$$

Step 2: Error Calculation

The predicted output is compared with the actual output. For binary classification, binary cross-entropy can be used:

$$L = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

Step 3: Backward Propagation

The error is propagated backward from the output layer to the hidden layer. Using the chain rule, gradients of the loss with respect to the weights and biases are calculated.

Step 4: Weight Update

The weights are updated using:

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

where (η) is the learning rate.

These steps are repeated for multiple training epochs until the error is minimized and the model learns to classify the patients correctly.

2. Actual and Predicted Outputs

Given dataset:

Example	Actual Output	Predicted Output
1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes
5	No	Yes

The predictions are compared with the actual outputs to construct the **confusion matrix**.

3. Confusion Matrix

For disease classification:

- **True Positive (TP):** Actual = Yes, Predicted = Yes
- **True Negative (TN):** Actual = No, Predicted = No
- **False Positive (FP):** Actual = No, Predicted = Yes
- **False Negative (FN):** Actual = Yes, Predicted = No

From the given dataset:

True Positives

Examples 1 and 4 are correctly predicted as Yes.

[
\\boxed{TP=2}
]

True Negatives

Example 3 is correctly predicted as No.

[
\\boxed{TN=1}
]

False Positives

Example 5 is actually No but is predicted as Yes.

[
\\boxed{FP=1}
]

False Negatives

Example 2 is actually Yes but is predicted as No.

[
\\boxed{FN=1}

4. Performance Evaluation

Accuracy

Accuracy represents the proportion of correctly classified samples.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$= \frac{2+1}{2+1+1+1}$$

$$= \frac{3}{5} = 0.60$$

$$\boxed{\text{Accuracy} = 60\%}$$

Precision

Precision indicates how many patients predicted as having the disease actually have the disease.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$= \frac{2}{2+1} \\ = \frac{2}{3}$$

$$\boxed{\text{Precision} = 66.67\%}$$

5. Role of Data Preprocessing

Data preprocessing improves the quality and reliability of the neural network.

Important preprocessing operations include:

1. **Handling missing values:** Missing Age, Blood Pressure, or Glucose values should be appropriately filled or handled.
2. **Encoding categorical data:** Values such as High, Normal, and Low can be converted into numerical representations.
3. **Feature scaling:** Numerical features such as Age and Glucose Level should be normalized or standardized so that features with larger numerical ranges do not dominate training.
4. **Noise and outlier handling:** Incorrect or abnormal measurements should be detected and treated appropriately.
5. **Data splitting:** Data should be divided into training, validation, and testing sets.

6. Feature Selection

Feature selection identifies the most relevant attributes for predicting the disease.

Potentially important features include:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

Methods such as **correlation analysis, mutual information, feature importance, and Recursive Feature Elimination (RFE)** can be used to identify useful features.

Removing irrelevant or redundant features can:

- Reduce model complexity.
- Reduce training time.
- Reduce overfitting.
- Improve generalization.
- Improve prediction reliability.

Conclusion

The neural network learns disease classification through **forward propagation, error calculation, backpropagation, and weight updates**. From the given predictions, the model produces **TP = 2, TN = 1, FP = 1, and FN = 1**, resulting in **60% accuracy, 66.67% precision, 66.67% recall, 50% specificity, and 66.67% F1-score**. Proper preprocessing and feature selection can further improve the model's accuracy, stability, and reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand